

ON CONJECTURAL FERMIONIC FORMULAS FOR THE MACDONALD INDEX IN ARGYRES–DOUGLAS THEORIES

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ABSTRACT. We prove a fermionic–bosonic duality relation for the Macdonald index in Argyres–Douglas theories of type (A_1, D_{2k+1}) , thereby yielding a conjectural fermionic formula due to Andrews et al. Our duality is built upon a new conjugate Bailey pair to be established using techniques from orthogonal polynomials and basic hypergeometric series. In addition, this fermionic formula implies another sum-like expression independently conjectured by Andrews et al. and Kim et al. for the same Macdonald index.

1. INTRODUCTION

Originating in [7], the *Argyres–Douglas theories* are families of four-dimensional $\mathcal{N} = 2$ superconformal field theories to describe the interactions of massless, mutually non-local BPS particles. Geometrically, they are realized by compactifying six-dimensional $\mathcal{N} = (2, 0)$ theories on a Riemann surface with irregular punctures [25, 26]. In particular, these theories have marginal deformations and the Coulomb branch spectrum is of fractional scaling dimension; see [8, 17, 18] for more background in physics and recent advances on this topic.

Due to the strong-coupling effects, no Lagrangian description exists in any duality frame for Argyres–Douglas theories, and there is a lack of *direct* ways for computing the superconformal indices, such as Schur index, Hall–Littlewood index, and Macdonald index. For the latter challenge, Buican and Nishinaka [12] made the very first progress on the Macdonald index in (A_1, A_{2k+1}) and (A_1, D_{2k}) Argyres–Douglas theories. In addition, they [13] proposed an Ansatz for the topological quantum description of Schur index, agreeing with the S -duality of Argyres–Douglas theories [11, 27], and this construction was subsequently shown to be consistent with S^1 reductions of Argyres–Douglas theories [14]. Alternatively, Córdova and Shao [15] predicted the Schur index according to the correspondence between superconformal field theories and vertex operator algebras, and their results match those in [10]. Along these directions, Song [24] made further headway on the computation of the three types of superconformal indices.

In this work, we focus on the Macdonald index in Argyres–Douglas theories of type (A_1, D_{2k+1}) . Briefly speaking, the *Macdonald index* $\mathcal{I}_{\text{Mac}}^{(2k+1)}(z, t; q)$, carrying fugacities z , t , and q , comes from a trace formula [21] for certain BPS states associated with the (A_1, D_{2k+1}) theory. This Macdonald index reduces to the Schur index at the $q = t$ case, and to the Hall–Littlewood index when $q = 0$.

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There is a *bosonic*¹ sum-like representation for $\mathcal{I}_{\text{Mac}}^{(2k+1)}(z, t; q)$ [20, p. 5, eq. (2.7)]:

$$\begin{aligned} \mathcal{I}_{\text{Mac}}^{(2k+1)}(z, t; q) &= \frac{1}{(t, tz^2, tz^{-2}; q)_{\infty}} \sum_{n \geq 0} (-1)^n t^{(k+1)n} q^{kn^2 + \binom{n}{2}} \frac{(q^{n+1}; q)_{\infty} (t^2 q^{2n}; q)_{\infty}}{(tq^n; q)_{\infty} (tq^{2n+1}; q)_{\infty}} \\ &\quad \times \sum_{j=0}^{2n} \frac{(t; q)_j (t; q)_{2n-j}}{(q; q)_j (q; q)_{2n-j}} z^{2j-2n}. \end{aligned} \quad (1.1)$$

Throughout, we adopt the conventional q -Pochhammer symbols:

$$(a; q)_{\infty} := \prod_{k \geq 0} (1 - aq^k), \quad (a; q)_n := \frac{(a; q)_{\infty}}{(aq^n; q)_{\infty}},$$

with the compact notation

$$\begin{aligned} (a_1, \dots, a_M; q)_{\infty} &:= (a_1; q)_{\infty} \cdots (a_M; q)_{\infty}, \\ (a_1, \dots, a_M; q)_n &:= (a_1; q)_n \cdots (a_M; q)_n. \end{aligned}$$

In addition, we need the q -binomial coefficients:

$$\begin{bmatrix} M \\ N \end{bmatrix}_q := \begin{cases} \frac{(q; q)_M}{(q; q)_N (q; q)_{M-N}}, & \text{if } 0 \leq N \leq M, \\ 0, & \text{otherwise.} \end{cases}$$

Recently, Andrews et al. [5], on the other hand, made the following conjecture² about the *fermionic* expression for $\mathcal{I}_{\text{Mac}}^{(2k+1)}(1, t; q)$.

Conjecture 1.1 (Andrews et al. [5, Conjecture 4]).

$$\mathcal{I}_{\text{Mac}}^{(2k+1)}(1, t; q) \stackrel{?}{=} \sum_{n_k \geq \dots \geq n_1 \geq 0} \frac{t^{n_1 + \dots + n_k} q^{n_1^2 + \dots + n_k^2}}{(q; q)_{n_k - n_{k-1}} \cdots (q; q)_{n_2 - n_1} (q; q)_{n_1}} \sum_{j=0}^{2n_k} \begin{bmatrix} 2n_k \\ j \end{bmatrix}_q. \quad (1.2)$$

Amongst basic hypergeometric series, especially (finite) Rogers–Ramanujan type identities [23], the *fermionic–bosonic duality* plays a significant role. In principle, it refers to equalities of the form

“multiple q -summation = single q -summation.”

A typical example is

$$\sum_{n_k \geq \dots \geq n_1 \geq 0} \frac{q^{n_1^2 + \dots + n_k^2}}{(q; q)_{n_k - n_{k-1}} \cdots (q; q)_{n_2 - n_1} (q; q)_{n_1}} = \frac{1}{(q; q)_{\infty}} \sum_{n=-\infty}^{\infty} (-1)^n q^{(k+1)n^2 + \binom{n}{2}},$$

which produces Andrews’ multiple generalization of the first Rogers–Ramanujan identity [1, p. 4082, Theorem 1] after applying Jacobi’s triple product [16, p. 357, eq. (II.28)] to the bosonic sum on the right-hand side.

¹Bosonic and fermionic representations are two ways to describe the same physical system: the former treats the system as a collection of independent particles, usually leading to single sum-like expressions mathematically, while the latter looks at the exclusion principle of the particles, resulting in multisums.

²The original formulation in [5, Conjecture 4] goes to (1.2) after a simple substitution of indices.

Now the first objective of this paper is to establish the following fermionic dual for (1.1):

$$\mathcal{I}_{\text{Mac}}^{(2k+1)}(z, t; q) = \sum_{n_k \geq \dots \geq n_1 \geq 0} \frac{t^{n_1 + \dots + n_k} q^{n_1^2 + \dots + n_{k-1}^2}}{(q; q)_{n_k - n_{k-1}} \cdots (q; q)_{n_2 - n_1} (q; q)_{n_1}} \sum_{j=0}^{2n_k} \begin{bmatrix} 2n_k \\ j \end{bmatrix}_q z^{2j-2n_k}, \quad (1.3)$$

which reduces to the conjectural formula of Andrews et al. [5, Conjecture 4], that is, (1.2), at $z = 1$. More precisely, we show the fermionic–bosonic duality:

Theorem 1.2. *For every $k \geq 1$,*

$$\begin{aligned} & \sum_{n_k \geq \dots \geq n_1 \geq 0} \frac{t^{n_1 + \dots + n_k} q^{n_1^2 + \dots + n_{k-1}^2}}{(q; q)_{n_k - n_{k-1}} \cdots (q; q)_{n_2 - n_1} (q; q)_{n_1}} \sum_{j=0}^{2n_k} \begin{bmatrix} 2n_k \\ j \end{bmatrix}_q z^{2j-2n_k} \\ &= \frac{1}{(t, tz^2, tz^{-2}; q)_\infty} \sum_{n \geq 0} (-1)^n t^{(k+1)n} q^{kn^2 + \binom{n}{2}} \frac{(q^{n+1}; q)_n (t^2 q^{2n}; q)_\infty}{(tq^n; q)_n (tq^{2n+1}; q)_\infty} \\ & \quad \times \sum_{j=0}^{2n} \frac{(t; q)_j (t; q)_{2n-j}}{(q; q)_j (q; q)_{2n-j}} z^{2j-2n}. \end{aligned} \quad (1.4)$$

Andrews et al. [6, eq. (26)] and Kim et al. [20, p. 8, eq. (2.26)] independently conjectured another fermionic sum for $\mathcal{I}_{\text{Mac}}^{(2k+1)}(z, t; q)$. Our second purpose is to show that this formula, as stated below using an equivalent form:³

$$\begin{aligned} \mathcal{I}_{\text{Mac}}^{(2k+1)}(z, t; q) &= (t, q; q)_\infty^k \sum_{\substack{r_1, \dots, r_k \geq 0 \\ s_1, \dots, s_k \geq 0}} \frac{t^{\sum_{i=1}^k s_i} q^{\sum_{i=1}^k r_i (s_{i-1} + s_i + 1)}}{(t, q; q)_{r_1} \cdots (t, q; q)_{r_k} (q; q)_{s_1}^2 \cdots (q; q)_{s_k}^2} \\ & \quad \times \sum_{u_1, u_2 \geq 0} \begin{bmatrix} s_k \\ u_1 \end{bmatrix}_q \begin{bmatrix} s_k \\ u_2 \end{bmatrix}_q z^{2u_1 - 2u_2}, \end{aligned} \quad (1.5)$$

with $s_0 := 0$, can be transformed to that in (1.3), thereby linking all three expressions for the Macdonald index in Argyres–Douglas theories of type (A_1, D_{2k+1}) . In other words, we prove the following identity:

Theorem 1.3. *For every $k \geq 1$,*

$$\begin{aligned} & \sum_{n_k \geq \dots \geq n_1 \geq 0} \frac{t^{n_1 + \dots + n_k} q^{n_1^2 + \dots + n_{k-1}^2}}{(q; q)_{n_k - n_{k-1}} \cdots (q; q)_{n_2 - n_1} (q; q)_{n_1}} \sum_{j=0}^{2n_k} \begin{bmatrix} 2n_k \\ j \end{bmatrix}_q z^{2j-2n_k} \\ &= (t, q; q)_\infty^k \sum_{\substack{r_1, \dots, r_k \geq 0 \\ s_1, \dots, s_k \geq 0}} \frac{t^{\sum_{i=1}^k s_i} q^{\sum_{i=1}^k r_i (s_{i-1} + s_i + 1)}}{(t, q; q)_{r_1} \cdots (t, q; q)_{r_k} (q; q)_{s_1}^2 \cdots (q; q)_{s_k}^2} \\ & \quad \times \sum_{u_1, u_2 \geq 0} \begin{bmatrix} s_k \\ u_1 \end{bmatrix}_q \begin{bmatrix} s_k \\ u_2 \end{bmatrix}_q z^{2u_1 - 2u_2}, \end{aligned} \quad (1.6)$$

where we put $s_0 := 0$.

³The equivalence can be seen by renaming the summation indices in [6, eq. (26)] or [20, p. 8, eq. (2.26)] and invoking the Cartan matrix of the D_{2k+1} Lie algebra. For details, see Appendix A.

This paper is structured as follows. First, in Section 2, we review Bailey's method on summations. Then a new conjugate Bailey pair is constructed in Section 3. In this course, techniques from orthogonal polynomials and basic hypergeometric series are needed, including a basic hypergeometric evaluation to be established in Appendix B. Next, we choose a proper seed Bailey pair in Section 4 to conclude a strengthening of Theorem 1.2. In Section 5, we prove Theorem 1.3. Finally, we provide a well-poised generalization of our new conjugate Bailey pair as closing remarks in Section 6.

2. SUMMATIONS À LA BAILEY

In this section, we briefly recall Bailey's theory on summations [9], which is streamlined in [2, Chapter 3]. Let $(u_n)_{n \geq 0}$ and $(v_n)_{n \geq 0}$ be generic sequences. Subject to suitable convergence conditions, we assume that sequences $(\alpha_n)_{n \geq 0}$, $(\beta_n)_{n \geq 0}$, $(\gamma_n)_{n \geq 0}$, and $(\delta_n)_{n \geq 0}$ satisfy the following relations for all $n \geq 0$:

$$\beta_n = \sum_{l=0}^n u_{n-l} v_{n+l} \alpha_l,$$

and

$$\gamma_n = \sum_{l \geq n} u_{l-n} v_{l+n} \delta_l.$$

Then *Bailey's transform* [2, p. 24, Theorem 3.1] asserts that

$$\sum_{n \geq 0} \alpha_n \gamma_n = \sum_{n \geq 0} \beta_n \delta_n. \quad (2.1)$$

A typical choice of the sequences u_n and v_n is

$$u_n = \frac{1}{(q; q)_n} \quad \text{and} \quad v_n = \frac{1}{(tq; q)_n},$$

where t is an indeterminate. We say sequences α_n and β_n form a *Bailey pair* relative to (t, q) [2, p. 25, eq. (3.27)] if for each $n \geq 0$,

$$\beta_n = \sum_{l=0}^n \frac{\alpha_l}{(q; q)_{n-l} (tq; q)_{n+l}}. \quad (2.2)$$

We may boost this Bailey pair to a *Bailey chain* [2, p. 29, eq. (3.38)]. To be precise, for each $k \geq 1$, the sequences

$$\alpha_n^{(k)} := \frac{(b_1^{-1}, c_1^{-1}, \dots, b_k^{-1}, c_k^{-1}; q)_n (b_1 c_1 \cdots b_k c_k)^n}{(b_1 tq, c_1 tq, \dots, b_k tq, c_k tq; q)_n} \cdot (tq)^{kn} \alpha_n \quad (2.3)$$

and

$$\begin{aligned} \beta_n^{(k)} := & \sum_{0 \leq n_0 \leq \dots \leq n_{k-1} \leq n} \frac{(tq)^{n_0 + \dots + n_{k-1}} \beta_{n_0}}{(q; q)_{n-n_{k-1}} (q; q)_{n_{k-1}-n_{k-2}} \cdots (q; q)_{n_1-n_0}} \\ & \times \frac{(b_k c_k tq; q)_{n-n_{k-1}} (b_{k-1} c_{k-1} tq; q)_{n_{k-1}-n_{k-2}} \cdots (b_1 c_1 tq; q)_{n_1-n_0}}{(b_1 tq, c_1 tq; q)_{n_1} \cdots (b_{k-1} tq, c_{k-1} tq; q)_{n_{k-1}} (b_k tq, c_k tq; q)_n} \\ & \times (b_1^{-1}, c_1^{-1}; q)_{n_0} \cdots (b_k^{-1}, c_k^{-1}; q)_{n_{k-1}} (b_1 c_1)^{n_0} \cdots (b_k c_k)^{n_{k-1}} \end{aligned} \quad (2.4)$$

also form a Bailey pair relative to (t, q) . This comes from iterations of *Bailey's lemma* [2, pp. 25–26, Theorem 3.3]. Finally, the *conjugate Bailey pair* (γ_n, δ_n) relative to (t, q) is such that

$$\gamma_n = \sum_{l \geq n} \frac{\delta_l}{(q; q)_{l-n} (tq; q)_{l+n}}. \quad (2.5)$$

Under different choices of Bailey and conjugate Bailey pairs, we are taken to a realm of summation formulas, with Bailey's transform recalled.

3. NEW CONJUGATE BAILEY PAIR

Conjugate Bailey pairs are not as well explored as Bailey pairs; see the work of Schilling and Warnaar [22] as an instance. Now our task is to establish the following new conjugate Bailey pair.

Theorem 3.1. *The sequences*

$$\gamma_n = \frac{t^n (q; q)_{2n} (t^2; q)_\infty}{(t^2; q)_{2n} (t, tq, tz, tz^{-1}; q)_\infty} \sum_{j=0}^{2n} \frac{(t; q)_j (t; q)_{2n-j}}{(q; q)_j (q; q)_{2n-j}} z^{j-n} \quad (3.1)$$

and

$$\delta_n = t^n \sum_{j=0}^{2n} \begin{bmatrix} 2n \\ j \end{bmatrix}_q z^{j-n} \quad (3.2)$$

form a conjugate Bailey pair relative to (t, q) .

Proof. Let $z := e^{i\theta}$. Recall the *continuous q -Hermite polynomials* [19, p. 320, eq. (13.1.7)] (or essentially the *Rogers–Szegő polynomials*; see [3, p. 49, Example 3]):

$$H_n(z; q) := \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix}_q z^{n-2j}, \quad (3.3)$$

and the *continuous q -ultraspherical polynomials* [19, p. 326, eq. (13.2.1)]:

$$C_n(z, t; q) := \sum_{j=0}^n \frac{(t; q)_j (t; q)_{n-j}}{(q; q)_j (q; q)_{n-j}} z^{n-2j}. \quad (3.4)$$

It is a standard result that these continuous q -Hermite polynomials form an orthogonal basis of symmetric Laurent polynomials in z . As such, we may assume that $c_{n,l}$ are coefficients in the infinite expansion of the symmetric bilateral even series

$$\frac{C_{2n}(z, t; q)}{(tz^2, tz^{-2}; q)_\infty} = \sum_{l \geq 0} c_{n,l} H_{2l}(z; q). \quad (3.5)$$

In particular, we do not have odd-indexed q -Hermite polynomials in this expansion because they are odd functions in z . Recall the *orthogonality* of continuous q -Hermite polynomials [19, p. 320, Theorem 13.1.3]:

$$\int_0^\pi H_n(z; q) H_{n'}(z; q) (z^2, z^{-2}; q)_\infty d\theta = \frac{2\pi (q; q)_n}{(q; q)_\infty} \delta_{n,n'},$$

where $\delta_{n,n'}$ is the *Kronecker delta*. Therefore,

$$c_{n,l} = \frac{(q; q)_\infty}{2(q; q)_{2l}} \int_0^\pi C_{2n}(z, t; q) H_{2l}(z; q) \frac{(z^2, z^{-2}; q)_\infty}{(tz^2, tz^{-2}; q)_\infty} \frac{d\theta}{\pi}.$$

Using (3.3) for $H_{2l}(z; q)$, we further write $c_{n,l}$ as

$$c_{n,l} = \frac{(q; q)_\infty}{2(q; q)_{2l}} \sum_{j=0}^{2l} \begin{bmatrix} 2l \\ j \end{bmatrix}_q \int_0^\pi z^{2l-2j} C_{2n}(z, t; q) \frac{(z^2, z^{-2}; q)_\infty}{(tz^2, tz^{-2}; q)_\infty} \frac{d\theta}{\pi}. \quad (3.6)$$

For the integral

$$I_{m,n} := \int_0^\pi z^{-2(m+n)} C_{2n}(z, t; q) \frac{(z^2, z^{-2}; q)_\infty}{(tz^2, tz^{-2}; q)_\infty} \frac{d\theta}{\pi}, \quad (3.7)$$

we apply (3.4) for $C_{2n}(z, t; q)$ and then invoke *Ramanujan's ${}_1\psi_1$ sum* [16, p. 357, eq. (II.29)]:

$$\sum_{k=-\infty}^{\infty} \frac{(a; q)_k z^k}{(b; q)_k} = \frac{(q, b/a, az, q/(az); q)_\infty}{(b, q/a, z, b/(az); q)_\infty}$$

with $(a, b, z) \mapsto (t^{-1}, t, tz^2)$ so that

$$\frac{(z^2, z^{-2}; q)_\infty}{(tz^2, tz^{-2}; q)_\infty} = \frac{(t, tq; q)_\infty (1 - z^{-2})}{(q, t^2; q)_\infty} \sum_{k=-\infty}^{\infty} \frac{(t^{-1}; q)_k t^k z^{2k}}{(t; q)_k}.$$

Hence, $I_{m,n}$ equals

$$\frac{(t, tq; q)_\infty}{(q, t^2; q)_\infty} \sum_{j=0}^{2n} \frac{(t; q)_j (t; q)_{2n-j}}{(q; q)_j (q; q)_{2n-j}} \sum_{k=-\infty}^{\infty} \frac{(t^{-1}; q)_k t^k}{(t; q)_k} \int_0^\pi z^{-2(m+j-k)} (1 - z^{-2}) \frac{d\theta}{\pi},$$

yielding the relation

$$I_{m,n} = \frac{(t, tq; q)_\infty}{(q, t^2; q)_\infty} (S_{m,n} - S_{m+1,n}), \quad (3.8)$$

where

$$S_{d,n} := \sum_{j=0}^{2n} \frac{(t; q)_j (t; q)_{2n-j} (t^{-1}; q)_{j+d} t^{j+d}}{(q; q)_j (q; q)_{2n-j} (t; q)_{j+d}}.$$

Now we write $S_{d,n}$ in terms of a ${}_3\phi_2$ series

$$S_{d,n} = \frac{(t; q)_{2n} (t^{-1}; q)_d t^d}{(q; q)_{2n} (t; q)_d} {}_3\phi_2 \left(\begin{matrix} t, t^{-1} q^d, q^{-2n} \\ tq^d, t^{-1} q^{1-2n} \end{matrix}; q, q \right),$$

and further evaluate it as

$$S_{d,n} = \frac{(t^2; q)_{2n} (q^d; q)_{2n} (t^{-1}; q)_d t^d}{(q; q)_{2n} (t; q)_{2n+d}}, \quad (3.9)$$

where we have applied the *q-Pfaff-Saalschütz sum* [16, p. 355, eq. (II.12)]:

$${}_3\phi_2 \left(\begin{matrix} a, b, q^{-n} \\ c, abq^{1-n}/c \end{matrix}; q, q \right) = \frac{(c/a, c/b; q)_n}{(c, c/(ab); q)_n}.$$

Recall from (3.6) and (3.8) that

$$\begin{aligned} c_{n,l} &= \frac{(q; q)_\infty}{2(q; q)_{2l}} \sum_{j=0}^{2l} \begin{bmatrix} 2l \\ j \end{bmatrix}_q I_{j-l-n,n} \\ &= \frac{(t, tq; q)_\infty}{2(t^2; q)_\infty (q; q)_{2l}} \sum_{j=0}^{2l} \begin{bmatrix} 2l \\ j \end{bmatrix}_q (S_{j-l-n,n} - S_{j-l-n+1,n}). \end{aligned}$$

For every j with $0 \leq j \leq 2l$, write $j' := 2l - j$. Our key observation is that for j in this range, it is always true that

$$\begin{bmatrix} 2l \\ j \end{bmatrix}_q S_{j-l-n,n} = - \begin{bmatrix} 2l \\ j' \end{bmatrix}_q S_{j'-l-n+1,n}.$$

Sum the above for j with $0 \leq j \leq 2l$ so that j' also runs from 0 to $2l$. Thus,

$$\sum_{j=0}^{2l} \begin{bmatrix} 2l \\ j \end{bmatrix}_q S_{j-l-n,n} = - \sum_{j=0}^{2l} \begin{bmatrix} 2l \\ j \end{bmatrix}_q S_{j-l-n+1,n}. \quad (3.10)$$

Consequently,

$$\begin{aligned} c_{n,l} &= \frac{(t, tq; q)_\infty}{(t^2; q)_\infty (q; q)_{2l}} \sum_{j=0}^{2l} \begin{bmatrix} 2l \\ j \end{bmatrix}_q S_{j-l-n,n} \\ &= \frac{(t, tq; q)_\infty (t^2; q)_{2n}}{(t^2; q)_\infty (q; q)_{2n}} \sum_{j=0}^{2l} \frac{(q^{j-l-n}; q)_{2n} (t^{-1}; q)_{j-l-n} t^{j-l-n}}{(q; q)_j (q; q)_{2l-j} (t; q)_{j-l+n}}, \end{aligned}$$

where (3.9) has been utilized. It then follows from (B.1) that

$$c_{n,l} = \frac{(t, tq; q)_\infty (t^2; q)_{2n} t^{l-n}}{(t^2; q)_\infty (q; q)_{2n} (q; q)_{l-n} (tq; q)_{l+n}}.$$

In particular,

$$c_{n,l} = 0, \quad (0 \leq l \leq n-1).$$

Finally, substituting these $c_{n,l}$ into (3.5), we have

$$\frac{t^n (q; q)_{2n} (t^2; q)_\infty}{(t^2; q)_{2n} (t, tq, tz^2, tz^{-2}; q)_\infty} C_{2n}(z, t; q) = \sum_{l \geq n} \frac{t^l}{(q; q)_{l-n} (tq; q)_{l+n}} H_{2l}(z; q).$$

Recalling (2.5), this exactly confirms the claimed conjugate Bailey pair under the change of variables $z \mapsto z^{-1/2}$. $\square$

Corollary 3.2. *For any Bailey pair (α_n, β_n) relative to (t, q) , we have*

$$\begin{aligned} &\sum_{n \geq 0} \beta_n \cdot t^n \sum_{j=0}^{2n} \begin{bmatrix} 2n \\ j \end{bmatrix}_q z^{j-n} \\ &= \frac{(t^2; q)_\infty}{(t, tq, tz, tz^{-1}; q)_\infty} \sum_{n \geq 0} \alpha_n \cdot \frac{t^n (q; q)_{2n}}{(t^2; q)_{2n}} \sum_{j=0}^{2n} \frac{(t; q)_j (t; q)_{2n-j}}{(q; q)_j (q; q)_{2n-j}} z^{j-n}. \end{aligned} \quad (3.11)$$

Proof. It follows from Bailey's transform (2.1) with the conjugate Bailey pair in Theorem 3.1 on call. $\square$

4. FERMIONIC–BOSONIC DUALITY

For the moment, we are ready to close our proof of the fermionic–bosonic duality in Theorem 1.2. Starting from the most elementary Bailey pair relative to (t, q) [2, p. 31, eqs. (3.47) and (3.48)]:

$$\alpha_n^{(0)} = \frac{(-1)^n q^{\binom{n}{2}} (1 - tq^{2n})(t; q)_n}{(1 - t)(q; q)_n}, \quad \beta_n^{(0)} = \delta_{n,0},$$

we produce a chain of Bailey pairs in light of (2.3) and (2.4):

$$\begin{aligned} \alpha_n^{(1)} &= \frac{(-1)^n t^n q^{n+\binom{n}{2}} (1 - tq^{2n})(t; q)_n}{(1 - t)(q; q)_n} \cdot \frac{(b_1^{-1}, c_1^{-1}; q)_n (b_1 c_1)^n}{(b_1 t q, c_1 t q; q)_n}, \\ \beta_n^{(1)} &= \frac{1}{(q; q)_n} \cdot \frac{(b_1 c_1 t q; q)_n}{(b_1 t q, c_1 t q; q)_n}, \end{aligned}$$

and for $k \geq 2$,

$$\begin{aligned} \alpha_n^{(k)} &= \frac{(-1)^n t^{kn} q^{kn+\binom{n}{2}} (1 - tq^{2n})(t; q)_n}{(1 - t)(q; q)_n} \\ &\quad \times \frac{(b_1^{-1}, c_1^{-1}, \dots, b_k^{-1}, c_k^{-1}; q)_n (b_1 c_1 \cdots b_k c_k)^n}{(b_1 t q, c_1 t q, \dots, b_k t q, c_k t q; q)_n}, \\ \beta_n^{(k)} &= \sum_{0 \leq n_1 \leq \dots \leq n_{k-1} \leq n} \frac{(tq)^{n_1 + \dots + n_{k-1}}}{(q; q)_{n-n_{k-1}} (q; q)_{n_{k-1}-n_{k-2}} \cdots (q; q)_{n_2-n_1} (q; q)_{n_1}} \\ &\quad \times \frac{(b_k c_k t q; q)_{n-n_{k-1}} (b_{k-1} c_{k-1} t q; q)_{n_{k-1}-n_{k-2}} \cdots (b_2 c_2 t q; q)_{n_2-n_1} (b_1 c_1 t q; q)_{n_1}}{(b_1 t q, c_1 t q; q)_{n_1} \cdots (b_{k-1} t q, c_{k-1} t q; q)_{n_{k-1}} (b_k t q, c_k t q; q)_n} \\ &\quad \times (b_2^{-1}, c_2^{-1}; q)_{n_1} \cdots (b_k^{-1}, c_k^{-1}; q)_{n_{k-1}} (b_2 c_2)^{n_1} \cdots (b_k c_k)^{n_{k-1}}. \end{aligned}$$

Apply them to (3.11) and make the substitution $z \mapsto z^2$. We immediately arrive at the following strengthening of Theorem 1.2, which reduces to (1.4) at the limiting case where all b 's and c 's go to zero.

Theorem 4.1. *For every $k \geq 1$,*

$$\begin{aligned} &\sum_{n_k \geq \dots \geq n_1 \geq 0} \frac{t^{n_1 + \dots + n_k} q^{n_1 + \dots + n_{k-1}}}{(q; q)_{n_k - n_{k-1}} \cdots (q; q)_{n_2 - n_1} (q; q)_{n_1}} \sum_{j=0}^{2n_k} \begin{bmatrix} 2n_k \\ j \end{bmatrix}_q z^{2j-2n_k} \\ &\quad \times \frac{(b_k c_k t q; q)_{n_k - n_{k-1}} \cdots (b_2 c_2 t q; q)_{n_2 - n_1} (b_1 c_1 t q; q)_{n_1}}{(b_1 t q, c_1 t q; q)_{n_1} \cdots (b_k t q, c_k t q; q)_{n_k}} \\ &\quad \times (b_2^{-1}, c_2^{-1}; q)_{n_1} \cdots (b_k^{-1}, c_k^{-1}; q)_{n_{k-1}} (b_2 c_2)^{n_1} \cdots (b_k c_k)^{n_{k-1}} \\ &= \frac{1}{(t, tz^2, tz^{-2}; q)_\infty} \sum_{n \geq 0} (-1)^n t^{(k+1)n} q^{kn+\binom{n}{2}} \frac{(q^{n+1}; q)_n (t^2 q^{2n}; q)_\infty}{(tq^n; q)_n (tq^{2n+1}; q)_\infty} \\ &\quad \times \sum_{j=0}^{2n} \frac{(t; q)_j (t; q)_{2n-j} z^{2j-2n}}{(q; q)_j (q; q)_{2n-j}} \cdot \frac{(b_1^{-1}, c_1^{-1}, \dots, b_k^{-1}, c_k^{-1}; q)_n (b_1 c_1 \cdots b_k c_k)^n}{(b_1 t q, c_1 t q, \dots, b_k t q, c_k t q; q)_n}. \end{aligned} \tag{4.1}$$

5. A SECOND FERMIONIC FORMULA

The goal of this section is to show the equivalence between (1.3) and (1.5) as stated in Theorem 1.3. To begin with, we single out summations over r 's on the right-hand side of (1.6). In particular, for each i with $1 \leq i \leq k$,

$$\begin{aligned} \sum_{r_i \geq 0} \frac{q^{r_i(s_{i-1}+s_i+1)}}{(t, q; q)_{r_i}} &= \lim_{\tau \rightarrow 0} {}_2\phi_1 \left(\begin{matrix} 0, \tau \\ t \end{matrix}; q, q^{s_{i-1}+s_i+1} \right) \\ &= \lim_{\tau \rightarrow 0} \frac{(\tau, 0; q)_\infty}{(t, q^{s_{i-1}+s_i+1}; q)_\infty} {}_2\phi_1 \left(\begin{matrix} t/\tau, q^{s_{i-1}+s_i+1} \\ 0 \end{matrix}; q, \tau \right), \end{aligned}$$

where we have applied *Heine's first transformation* [16, p. 359, eq. (III.1)]:

$${}_2\phi_1 \left(\begin{matrix} a, b \\ c \end{matrix}; q, z \right) = \frac{(b, az; q)_\infty}{(c, z; q)_\infty} {}_2\phi_1 \left(\begin{matrix} c/b, z \\ az \end{matrix}; q, b \right).$$

Thus,

$$(t, q; q)_\infty \sum_{r_i \geq 0} \frac{q^{r_i(s_{i-1}+s_i+1)}}{(t, q; q)_{r_i}} = \sum_{r_i \geq 0} \frac{(-1)^{r_i} t^{r_i} q^{\binom{r_i}{2}} (q; q)_{s_{i-1}+s_i+r_i}}{(q; q)_{r_i}}.$$

Before substituting the above relations into the right-hand side of (1.6), we change the indices for each i with $1 \leq i \leq k$,

$$n_i := r_i + s_i.$$

It follows that

$$\text{RHS (1.6)} = \sum_{n_1, \dots, n_k \geq 0} t^{\sum_{i=1}^k n_i} (q; q)_{n_1} B_{n_k}(z; q) \prod_{i=1}^{k-1} \Phi_{n_i, n_{i+1}}(q), \quad (5.1)$$

where

$$B_n(z; q) := \sum_{s=0}^n \frac{(-1)^{n-s} q^{\binom{n-s}{2}}}{(q; q)_s^2 (q; q)_{n-s}} \sum_{u_1, u_2 \geq 0} \begin{bmatrix} s \\ u_1 \end{bmatrix}_q \begin{bmatrix} s \\ u_2 \end{bmatrix}_q z^{2u_1-2u_2},$$

and

$$\Phi_{n, n'}(q) := \sum_{s=0}^n \frac{(-1)^{n-s} q^{\binom{n-s}{2}} (q; q)_{s+n'}}{(q; q)_s^2 (q; q)_{n-s}}.$$

For $B_n(z; q)$, we write it in terms of the continuous q -Hermite polynomial (3.3):

$$B_n(z; q) = \sum_{s=0}^n \frac{(-1)^{n-s} q^{\binom{n-s}{2}}}{(q; q)_s^2 (q; q)_{n-s}} H_s(z; q)^2,$$

where we have noted the symmetry $H_n(z; q) = H_n(z^{-1}; q)$. Now we require the linearization of products of $H_n(z; q)$ [19, p. 322, Theorem 13.1.5]:

$$H_m(z; q) H_n(z; q) = \sum_{l=0}^{\min(m, n)} \frac{(q; q)_m (q; q)_n}{(q; q)_l (q; q)_{m-l} (q; q)_{n-l}} H_{m+n-2l}(z; q).$$

It follows that

$$B_n(z; q) = \sum_{s=0}^n \frac{(-1)^{n-s} q^{\binom{n-s}{2}}}{(q; q)_s^2 (q; q)_{n-s}} \sum_{l=0}^s \frac{(q; q)_s^2}{(q; q)_l (q; q)_{s-l}} H_{2s-2l}(z; q).$$

Substituting $l \mapsto s - j$ and interchanging the summations, we have

$$B_n(z; q) = \sum_{j=0}^n \frac{H_{2j}(z; q)}{(q; q)_j^2} \sum_{s=j}^n \frac{(-1)^{n-s} q^{\binom{n-s}{2}}}{(q; q)_{n-s} (q; q)_{s-j}}.$$

For the inner sum, we further make the change of indices $s \mapsto s + j$ and write it as a ${}_1\phi_0$ series so that

$$B_n(z; q) = \sum_{j=0}^n \frac{(-1)^{n-j} q^{\binom{n}{2} + \binom{j}{2} + j(1-n)} H_{2j}(z; q)}{(q; q)_j^2} {}_1\phi_0 \left(\begin{matrix} q^{-(n-j)} \\ - \end{matrix}; q, q \right),$$

while this ${}_1\phi_0$ series becomes the Kronecker delta $\delta_{j,n}$ according to the *q-binomial theorem* [16, p. 354, eq. (II.4)]:

$${}_1\phi_0 \left(\begin{matrix} q^{-n} \\ - \end{matrix}; q, z \right) = (zq^{-n}; q)_n.$$

Therefore, by further using the symmetry $H_n(z; q) = H_n(z^{-1}; q)$,

$$B_n(z; q) = \frac{H_{2n}(z; q)}{(q; q)_n^2} = \frac{1}{(q; q)_n^2} \sum_{j=0}^{2n} \begin{bmatrix} 2n \\ j \end{bmatrix}_q z^{2j-2n}. \quad (5.2)$$

For $\Phi_{n,n'}(q)$, we have

$$\begin{aligned} \Phi_{n,n'}(q) &= \frac{(-1)^n q^{\binom{n}{2}} (q; q)_{n'}}{(q; q)_n} {}_2\phi_1 \left(\begin{matrix} q^{n'+1}, q^{-n} \\ q \end{matrix}; q, q \right) \\ &= \frac{(-1)^n q^{\binom{n}{2}} (q; q)_{n'}}{(q; q)_n} \cdot \frac{q^{n(n'+1)} (q^{-n'}; q)_n}{(q; q)_n}, \end{aligned}$$

where for the evaluation of the ${}_2\phi_1$ series, we have used the *second q-Chu-Vandermonde sum* [16, p. 354, eq. (II.6)]:

$${}_2\phi_1 \left(\begin{matrix} a, q^{-n} \\ c \end{matrix}; q, q \right) = \frac{a^n (c/a; q)_n}{(c; q)_n}.$$

Hence,

$$\Phi_{n,n'}(q) = \frac{q^{n^2} (q; q)_{n'}}{(q; q)_n} \begin{bmatrix} n' \\ n \end{bmatrix}_q. \quad (5.3)$$

Finally, applying (5.2) and (5.3) to (5.1), we find that

$$\text{RHS (1.6)} = \sum_{n_k \geq \dots \geq n_1 \geq 0} \frac{t^{n_1 + \dots + n_k} q^{n_1^2 + \dots + n_{k-1}^2}}{(q; q)_{n_k - n_{k-1}} \cdots (q; q)_{n_2 - n_1} (q; q)_{n_1}} \sum_{j=0}^{2n_k} \begin{bmatrix} 2n_k \\ j \end{bmatrix}_q z^{2j-2n_k},$$

as claimed.

6. CLOSING REMARKS

In [4], Andrews extended the scheme of Bailey pairs to a well-poised version. To be specific, sequences α'_n and β'_n form a *WP-Bailey pair* relative to (s, t, q) [4, p. 15, eq. (6.2)] if for each $n \geq 0$,

$$\beta'_n = \sum_{l=0}^n \frac{(st^{-1}; q)_{n-l} (s; q)_{n+l}}{(q; q)_{n-l} (tq; q)_{n+l}} \alpha'_l; \quad (6.1)$$

likewise, sequences γ'_n and δ'_n form a *conjugate WP-Bailey pair* relative to (s, t, q) if for each $n \geq 0$,

$$\gamma'_n = \sum_{l \geq n} \frac{(st^{-1}; q)_{l-n} (s; q)_{l+n}}{(q; q)_{l-n} (tq; q)_{l+n}} \delta'_l. \quad (6.2)$$

It is notable that such a well-poised extension reduces to the original form by taking $s = 0$.

Toward this direction, we may lift the conjugate Bailey pair in Theorem 3.1 to the following WP-generalization, which will be established in Appendix C. However, it remains unknown to understand the physical meaning conveyed by this pair, especially the new parameter s .

Theorem 6.1. *The sequences*

$$\gamma'_n = \frac{t^n (q; q)_{2n} (t^2, sz, sz^{-1}; q)_\infty}{(t^2; q)_{2n} (t, tq, tz, tz^{-1}; q)_\infty} \sum_{j=0}^{2n} \frac{(t; q)_j (t; q)_{2n-j}}{(q; q)_j (q; q)_{2n-j}} z^{j-n} \quad (6.3)$$

and

$$\delta'_n = \frac{t^n (1 - sq^{2n}) (q; q)_{2n} (s^2; q)_\infty}{(1 - s) (s^2; q)_{2n} (s, sq; q)_\infty} \sum_{j=0}^{2n} \frac{(s; q)_j (s; q)_{2n-j}}{(q; q)_j (q; q)_{2n-j}} z^{j-n} \quad (6.4)$$

form a *conjugate WP-Bailey pair* relative to (s, t, q) .

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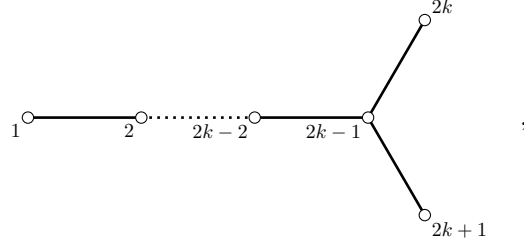
APPENDIX A. ORIGINAL FORMULATION OF THE FERMIONIC SUM IN (1.5)

Here we examine the equivalence between the fermionic sum in (1.5) and that in Andrews et al. [6] or Kim et al. [20]. Note that [20, p. 8, eq. (2.26)], with z rescaled to z^2 and T substituted by tq^{-1} , reads:

$$\begin{aligned}
 (t, q; q)_\infty^k & \sum_{\substack{l_1, \dots, l_{2k+1} \geq 0 \\ m_1, \dots, m_{2k+1} \geq 0}} \delta_{l_{2k} + l_{2k+1}, m_{2k} + m_{2k+1}} \prod_{i=1}^{2k-1} \delta_{l_i, m_i} \\
 & \times \frac{q^{\sum_{i,j=1}^{2k+1} \frac{a_{i,j} l_i m_j}{2} + \sum_{i=1}^k \frac{l_{2i-1} + m_{2i-1}}{2}} t^{\sum_{i=1}^k \frac{l_{2i} + m_{2i}}{2} + \frac{l_{2k+1} + m_{2k+1}}{2}} z^{2m_{2k+1} - 2l_{2k+1}}}{(q; q)_{l_{2k+1}} (q; q)_{m_{2k+1}} \prod_{i=1}^k (q; q)_{l_{2i}} (q; q)_{m_{2i}} (t; q)_{l_{2i-1}} (q; q)_{m_{2i-1}}}, \quad (\text{A.1})
 \end{aligned}$$

where $A := (a_{i,j})_{1 \leq i,j \leq 2k+1}$ is given by $A = 2I_{2k+1} - M_{\text{Car}}(D_{2k+1})$ with I_{2k+1} the identity matrix of dimension $2k+1$ and $M_{\text{Car}}(D_{2k+1})$ the Cartan matrix of the

D_{2k+1} Lie algebra. In particular, in this case A is the adjacency matrix for the D_{2k+1} Dynkin diagram:



so that $a_{i,i+1} = a_{i+1,i} = 1$ for i with $1 \leq i \leq 2k-2$, $a_{2k-1,2k} = a_{2k,2k-1} = a_{2k-1,2k+1} = a_{2k+1,2k-1} = 1$, and $a_{i,j} = 0$ for all other cases.

In light of the Kronecker delta factors in (A.1), we rename the summation indices as in (1.5):

$$\begin{aligned} r_i &:= l_{2i-1} = m_{2i-1}, & (1 \leq i \leq k), \\ s_i &:= l_{2i} = m_{2i}, & (1 \leq i \leq k-1), \end{aligned}$$

and

$$s_k := l_{2k} + l_{2k+1} = m_{2k} + m_{2k+1}.$$

Thus, the exponent of q in the numerator becomes

$$\sum_{i,j=1}^{2k+1} \frac{a_{i,j} l_i m_j}{2} + \sum_{i=1}^k \frac{l_{2i-1} + m_{2i-1}}{2} = \sum_{i=1}^k r_i (s_{i-1} + s_i + 1),$$

while the exponent of t is

$$\sum_{i=1}^k \frac{l_{2i} + m_{2i}}{2} + \frac{l_{2k+1} + m_{2k+1}}{2} = \sum_{i=1}^k s_i.$$

We further put

$$u_1 := l_{2k}, \quad u_2 := m_{2k},$$

so that

$$l_{2k+1} = s_k - u_1, \quad m_{2k+1} = s_k - u_2.$$

In this way, the exponent of z becomes

$$2m_{2k+1} - 2l_{2k+1} = 2u_1 - 2u_2.$$

Finally, for the denominator, we have

$$\frac{1}{(t, q; q)_{r_1} \cdots (t, q; q)_{r_k} (q; q)_{s_1}^2 \cdots (q; q)_{s_{k-1}}^2} \cdot \frac{1}{(q; q)_{l_{2k}} (q; q)_{m_{2k}} (q; q)_{l_{2k+1}} (q; q)_{m_{2k+1}}},$$

while the latter factor can be further rewritten as

$$\frac{1}{(q; q)_{l_{2k} + l_{2k+1}} (q; q)_{m_{2k} + m_{2k+1}}} \begin{bmatrix} l_{2k} + l_{2k+1} \\ l_{2k} \end{bmatrix}_q \begin{bmatrix} m_{2k} + m_{2k+1} \\ m_{2k} \end{bmatrix}_q = \frac{1}{(q; q)_{s_k}^2} \begin{bmatrix} s_k \\ u_1 \end{bmatrix}_q \begin{bmatrix} s_k \\ u_2 \end{bmatrix}_q.$$

Now (A.1) becomes exactly the same as the fermionic sum in (1.5).

APPENDIX B. A BASIC HYPERGEOMETRIC IDENTITY

Here we evaluate the following basic hypergeometric series, which appears as a critical step in our proof of Theorem 3.1.

Lemma B.1. *For $l, n \geq 0$,*

$$\sum_{j=0}^{2l} \frac{(q^{j-l-n}; q)_{2n} (t^{-1}; q)_{j-l-n} t^j}{(q; q)_j (q; q)_{2l-j} (t; q)_{j-l+n}} = \frac{t^{2l}}{(q; q)_{l-n} (tq; q)_{l+n}}. \quad (\text{B.1})$$

Proof. If $l < n$, we have the vanishing of $(q^{j-l-n}; q)_{2n}$ for all j ranging from 0 to $2l$, and hence both sides of (B.1) trivially go to zero in this case. Now we assume $l \geq n$. For convenience, let us write

$$s_j := \frac{(q^{j-l-n}; q)_{2n} (t^{-1}; q)_{j-l-n} t^j}{(q; q)_j (q; q)_{2l-j} (t; q)_{j-l+n}}, \quad (0 \leq j \leq 2l).$$

It is clear that s_j is supported on the ranges $0 \leq j \leq l-n$ and $l+n+1 \leq j \leq 2l$. Putting in addition

$$s_{2l+1} := 0,$$

we observe that for each j with $0 \leq j \leq l-n$,

$$s_j + s_{1+2l-j} = \frac{q^j (1 - q^{1+2l-2j})}{1 - q^{1+2l-j}} \cdot s_j.$$

Thus,

$$\text{LHS (B.1)} = \sum_{j=0}^{l-n} \frac{(q^{j-l-n}; q)_{2n} (t^{-1}; q)_{j-l-n} t^j q^j (1 - q^{1+2l-2j})}{(q; q)_j (q; q)_{2l-j} (t; q)_{j-l+n} (1 - q^{1+2l-j})}.$$

Note that

$$(q^{j-l-n}; q)_{2n} = \frac{(q; q)_{l+n-j} q^{n(2j-2l-1)}}{(q; q)_{l-n-j}}.$$

We further make the change of indices $j \mapsto l-n-j$ so that

$$\begin{aligned} \text{LHS (B.1)} &= \sum_{j=0}^{l-n} \frac{(q; q)_{2n+j} (t^{-1}; q)_{-2n-j} t^{l-n-j} q^{l-2n(n+1)-(2n+1)j} (1 - q^{1+2n+2j})}{(q; q)_j (q; q)_{l-n-j} (q; q)_{l+n+j} (t; q)_{-j} (1 - q^{1+l+n+j})} \\ &= \sum_{j=0}^{l-n} \frac{(q; q)_{2n+j} (t^{-1}; q)_{-2n-j} t^{l-n-j} q^{l-2n(n+1)-(2n+1)j} (1 - q^{1+2n+2j})}{(q; q)_j (q; q)_{l-n-j} (q; q)_{1+l+n+j} (t; q)_{-j}}. \end{aligned}$$

Using the relations

$$\begin{aligned} (a; q)_{m+j} &= (a; q)_m (aq^m; q)_j, \\ (a; q)_{m-j} &= \frac{(a; q)_m}{(q^{1-m}/a; q)_j} \left(-\frac{q}{a}\right)^j q^{\binom{j}{2}-nj}, \end{aligned}$$

we have

$$\begin{aligned} \text{LHS (B.1)} &= \frac{(q; q)_{2n} t^{l+n} q^{l-n}}{(q; q)_{l-n} (q; q)_{l+n+1} (tq; q)_{2n}} \\ &\times \sum_{j=0}^{l-n} \frac{(q^{2n+1}, t^{-1}q, q^{-(l-n)}; q)_j t^j q^{(l-n-1)j} (1 - q^{1+2n+2j})}{(q, q^{l+n+2}, tq^{2n+1}; q)_j (-1)^j q^{\binom{j}{2}}} \end{aligned}$$

$$\begin{aligned}
 &= \frac{(q; q)_{2n} t^{l+n} q^{l-n} (1 - q^{2n+1})}{(q; q)_{l-n} (q; q)_{l+n+1} (tq; q)_{2n}} \\
 &\quad \times \lim_{\tau \rightarrow 0} {}_6\phi_5 \left(\begin{matrix} q^{2n+1}, q^{n+\frac{3}{2}}, -q^{n+\frac{3}{2}}, t^{-1}q, \tau q^{2n+2}, q^{-(l-n)} \\ q^{n+\frac{1}{2}}, -q^{n+\frac{1}{2}}, tq^{2n+1}, \tau^{-1}, q^{l+n+2} \end{matrix}; q, \frac{tq^{l-n-1}}{\tau} \right).
 \end{aligned}$$

For this terminating ${}_6\phi_5$ series, we require the following evaluation [16, p. 356, eq. (II.21)]:

$${}_6\phi_5 \left(\begin{matrix} a, a^{\frac{1}{2}}q, -a^{\frac{1}{2}}q, b, c, q^{-n} \\ a^{\frac{1}{2}}, -a^{\frac{1}{2}}, aq/b, aq/c, aq^{n+1} \end{matrix}; q, \frac{aq^{n+1}}{bc} \right) = \frac{(aq, aq/(bc); q)_n}{(aq/b, aq/c; q)_n}. \quad (\text{B.2})$$

Therefore,

$$\begin{aligned}
 \text{LHS (B.1)} &= \frac{(q; q)_{2n} t^{l+n} q^{l-n} (1 - q^{2n+1})}{(q; q)_{l-n} (q; q)_{l+n+1} (tq; q)_{2n}} \lim_{\tau \rightarrow 0} \frac{(q^{2n+2}, tq^{-1}\tau^{-1}; q)_{l-n}}{(tq^{2n+1}, \tau^{-1}; q)_{l-n}} \\
 &= \frac{t^{l+n} q^{l-n}}{(q; q)_{l-n} (tq; q)_{l+n}} \lim_{\tau \rightarrow 0} \frac{(tq^{-1}\tau^{-1}; q)_{l-n}}{(\tau^{-1}; q)_{l-n}}.
 \end{aligned}$$

Finally, using the relation

$$\frac{(a; q)_n}{(b; q)_n} = \frac{(q^{1-n}/a; q)_n}{(q^{1-n}/b; q)_n} \left(\frac{a}{b} \right)^n,$$

the above limit becomes $(tq^{-1})^{l-n}$, producing the right-hand side of (B.1). $\square$

APPENDIX C. LIFTING TO A CONJUGATE WP-BAILEY PAIR

Here we show the conjugate WP-Bailey pair claimed in Theorem 6.1; the proof is similar to that for Theorem 3.1. In particular, this time we use the fact that the continuous q -ultraspherical polynomials $C_n(z, s; q)$ also form an orthogonal basis of symmetric Laurent polynomials in z . Then it is possible to write

$$\frac{(sz^2, sz^{-2}; q)_\infty}{(tz^2, tz^{-2}; q)_\infty} C_{2n}(z, t; q) = \sum_{l \geq 0} c'_{n,l} C_{2l}(z, s; q).$$

Applying the orthogonality of continuous q -ultraspherical polynomials [19, p. 327, Theorem 13.2.1]:

$$\int_0^\pi C_n(z, s; q) C_{n'}(z, s; q) \frac{(z^2, z^{-2}; q)_\infty}{(sz^2, sz^{-2}; q)_\infty} d\theta = \frac{2\pi(1-s)(s^2; q)_n (s, sq; q)_\infty}{(1-sq^n)(q; q)_n (q, s^2; q)_\infty} \delta_{n,n'},$$

the coefficients $c'_{n,l}$ can be identified by

$$\begin{aligned}
 c'_{n,l} &= \frac{(1-sq^{2l})(q; q)_{2l} (q, s^2; q)_\infty}{2(1-s)(s^2; q)_{2l} (s, sq; q)_\infty} \int_0^\pi C_{2n}(z, t; q) C_{2l}(z, s; q) \frac{(z^2, z^{-2}; q)_\infty}{(tz^2, tz^{-2}; q)_\infty} \frac{d\theta}{\pi} \\
 &= \frac{(1-sq^{2l})(q; q)_{2l} (q, s^2; q)_\infty}{2(1-s)(s^2; q)_{2l} (s, sq; q)_\infty} \sum_{j=0}^{2l} \frac{(s; q)_j (s; q)_{2l-j}}{(q; q)_j (q; q)_{2l-j}} I_{j-l-n,n},
 \end{aligned}$$

where $I_{j-l-n,n}$ is as in (3.7). It follows from (3.8) and (3.9), with a similar symmetry to that in (3.10) used, that

$$c'_{n,l} = \frac{t^{-l-n} (1-sq^{2l})(q; q)_{2l} (t^2; q)_{2n} (t, tq, s^2; q)_\infty}{(1-s)(s^2; q)_{2l} (q; q)_{2n} (t^2, s, sq; q)_\infty}$$

$$\times \sum_{j=0}^{2l} \frac{(s; q)_j (s; q)_{2l-j} (q^{j-l-n}; q)_{2n} (t^{-1}; q)_{j-l-n} t^j}{(q; q)_j (q; q)_{2l-j} (t; q)_{j-l+n}}.$$

For the summation on the right-hand side, we proceed in the same way as that for Lemma B.1 by pairing the summands, and eventually reformulate it in terms of a ${}_6\phi_5$ series:

$$\begin{aligned} & \sum_{j=0}^{2l} \frac{(s; q)_j (s; q)_{2l-j} (q^{j-l-n}; q)_{2n} (t^{-1}; q)_{j-l-n} t^j}{(q; q)_j (q; q)_{2l-j} (t; q)_{j-l+n}} \\ &= \frac{(s; q)_{l-n} (s; q)_{l+n} (q; q)_{2n} t^{l+n} (1 - s^{-1}q) (1 - q^{2n+1})}{(q; q)_{l-n} (q; q)_{l+n+1} (tq; q)_{2n} (1 - s^{-1}q^{1-(l-n)})} \\ & \quad \times {}_6\phi_5 \left(\begin{matrix} q^{2n+1}, q^{n+\frac{3}{2}}, -q^{n+\frac{3}{2}}, t^{-1}q, sq^{l+n}, q^{-(l-n)} \\ q^{n+\frac{1}{2}}, -q^{n+\frac{1}{2}}, tq^{2n+1}, s^{-1}q^{2-(l-n)}, q^{l+n+2}; q, \frac{tq}{s} \end{matrix} \right). \end{aligned}$$

Invoking (B.2), we arrive at

$$\sum_{j=0}^{2l} \frac{(s; q)_j (s; q)_{2l-j} (q^{j-l-n}; q)_{2n} (t^{-1}; q)_{j-l-n} t^j}{(q; q)_j (q; q)_{2l-j} (t; q)_{j-l+n}} = \frac{(st^{-1}; q)_{l-n} (s; q)_{l+n} t^{2l}}{(q; q)_{l-n} (tq; q)_{l+n}}. \quad (\text{C.1})$$

Substituting (C.1) into the expression for $c'_{n,l}$, we conclude that

$$c'_{n,l} = \frac{(st^{-1}; q)_{l-n} (s; q)_{l+n}}{(q; q)_{l-n} (tq; q)_{l+n}} \cdot \frac{t^{l-n} (1 - sq^{2l}) (q; q)_{2l} (t^2; q)_{2n} (t, tq, s^2; q)_{\infty}}{(1 - s) (s^2; q)_{2l} (q; q)_{2n} (t^2, s, sq; q)_{\infty}},$$

which, in particular, vanishes when $l < n$. Therefore,

$$\begin{aligned} & \frac{t^n (q; q)_{2n} (t^2, sz^2, sz^{-2}; q)_{\infty}}{(t^2; q)_{2n} (t, tq, tz^2, tz^{-2}; q)_{\infty}} C_{2n}(z, t; q) \\ &= \sum_{l \geq n} \frac{(st^{-1}; q)_{l-n} (s; q)_{l+n}}{(q; q)_{l-n} (tq; q)_{l+n}} \cdot \frac{t^l (1 - sq^{2l}) (q; q)_{2l} (s^2; q)_{\infty}}{(1 - s) (s^2; q)_{2l} (s, sq; q)_{\infty}} C_{2l}(z, s; q). \end{aligned}$$

Finally, we replace z with $z^{-1/2}$. The above identity becomes exactly the required relation (6.2) for conjugate WP-Bailey pairs.

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